

Kubernetes learning path

Broad map, not a syllabus. One spine project. One heading at a time.

How we work

You ask "what do I learn now?". We open a single heading, apply it on the Library app (internship project: API + UI on Minikube), then we move. No lecture dump until that heading is open.

Spine project

Keep Library. Each heading upgrades that deploy - Compose, cluster YAML, Helm, then GitOps - instead of a new toy repo every week.

Phases

A - Docker

Images, data, networks, Compose, ship to a host.

Done when: Library runs locally with Compose.

B - Kubernetes apps

Workloads, config, probes, services, ingress.

Done when: You can explain and change the existing Minikube deploy on purpose.

C - Workloads and storage

Jobs, StatefulSets, volumes, PV/PVC.

Done when: You know when a Deployment is the wrong tool, and data can outlive a pod.

D - Helm

Packaging.

Done when: Library installs and upgrades from a chart and values, not a folder of apply commands.

E - Operate

Debug, metrics, logs.

Done when: You find a broken pod from events, logs, and metrics - not from guessing YAML.

F - Delivery

CI image build and GitOps (Argo CD or Flux).

Done when: A git change is what the cluster should look like.

G - Security

RBAC, policies, secrets hygiene.

Done when: Least privilege on Library; secrets are not treated as ConfigMaps.

H - Clusters

kind/kubeadm, nodes, scheduling.

Done when: You can reason about the cluster, not only the app.

I - Managed Kubernetes

One of AKS, EKS, or GKE.

Done when: You can describe how Library would sit on a cloud cluster.

Out of scope until later

Service mesh, writing operators, Terraform-everything, CKS cram. Optional after H/I.

Current heading

Phase A - Docker: images and containers. Closed until you say we start. Then: Dockerfile, image vs container, run/stop/logs - on Library images.